

Arduino-Based 8 LED Control System Using Push Button Switch

Abstract

This project presents the design and simulation of an Arduino Uno-based LED control system using a push button switch. The system controls eight LEDs simultaneously through a single input signal. The project demonstrates fundamental embedded system concepts such as digital input sensing, digital output control, and real-time hardware interfacing. The complete circuit is designed and tested using Proteus ISIS Professional, and the control logic is implemented using Embedded C.

Introduction

Embedded systems are widely used in automation, control, and monitoring applications. Arduino Uno is a popular microcontroller platform due to its simplicity and ease of use. This project introduces basic digital input-output interfacing by controlling multiple LEDs using a single push button, making it ideal for beginners.

Problem Statement

Controlling multiple output indicators individually increases circuit complexity. A centralized and simple control mechanism is required to operate multiple output devices efficiently using minimal hardware.

Objective

The objective of this project is to design and implement a simple embedded system that controls eight LEDs using a single push button switch interfaced with an Arduino Uno.

System Architecture

The system consists of an input unit (push button), a processing unit (Arduino Uno), and an output unit (LEDs). The Arduino continuously monitors the switch status and controls the LEDs based on the input signal.

Hardware Components

- Arduino Uno (ATmega328P)
- Red LEDs – 8 numbers
- 330Ω resistors – 8 numbers
- Push button switch
- 10kΩ pull-down resistor
- Connecting wires

Software Used

- Arduino IDE for Embedded C programming
- Proteus ISIS Professional for simulation

Pin Configuration

- Digital pins D13 to D6 connected to LEDs
- Analog pin A0 connected to push button
- GND used as common ground

Circuit Diagram Description

- Arduino Uno acts as the main controller

- Each LED is connected through a 330Ω current-limiting resistor
- LED cathodes are connected to ground
- Push button is connected to analog pin A0
- Pull-down resistor ensures stable input
- Circuit is simulated using Proteus ISIS Professional

Working Principle

- Arduino reads the status of the push button continuously
- When switch is pressed, input becomes HIGH
- All eight LEDs turn ON
- When switch is released, input becomes LOW
- All LEDs turn OFF

Results

The system successfully controls all eight LEDs using a single push button. The LEDs respond accurately to the switch input, demonstrating reliable digital input sensing and output control in real time.

Advantages

- Simple and easy to understand design
- Low-cost components
- Real-time operation
- Beginner-friendly embedded project

Real-Time Applications

- Industrial indicator panels
- Emergency alert lighting systems
- Home automation lighting control
- Educational embedded system training kits

Future Scope

The project can be extended by integrating wireless modules such as Bluetooth or Wi-Fi for remote control. Individual LED control, mobile application integration, and power optimization techniques can further enhance the system.

Conclusion

This project successfully demonstrates fundamental embedded system concepts using Arduino Uno. It provides a strong foundation for advanced embedded and automation projects.